Yiwen Chen

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Education

University of Chicago Sep. 2026 – Present
M.S. in Data Science Chicago, IL

University of Minnesota, Twin Cities Sep. 2023 – May 2026
B.S. in Statistics; Minor in Mathematics Minneapolis, MN
GPA: 3.9/4.0

Southwest University of Political Science and Law Sep. 2021 – Jun. 2023
Economic Statistics (transferred) Chongqing, China

Research Interests

Computational social science; educational inequality and opportunity; family contexts and adolescent development; social determinants of health; statistical modeling; interpretable machine learning.

Research Experience

Technology, Teens, and Families Lab, University of Minnesota Sep. 2024 – 2026
Undergraduate Research Assistant, advised by Prof. Xiaoran Sun Minneapolis, MN

- Studied how adolescents' family experiences relate to educational achievement using Add Health data, applying unsupervised learning to identify distinct family-experience profiles and link them to educational outcomes.
- Contributed across the research process: developing research questions and study design, constructing variables in R, handling missing data, clustering analysis, interpreting results, and drafting the manuscript.
- Conducted a literature review on overnight smartphone use, summarizing measurement approaches, statistical methods, and key empirical findings to inform the team's research plans.

Oak Ridge National Laboratory May 2025 – Aug. 2025
Pathways to Computing Internship Program (PCIP), advised by Dr. Seung-Hwan Lim and Dr. Guojing Cong Oak Ridge, TN

- Characterized cross-species toxicogenomic data through exploratory analysis of human and rat transcriptomic datasets (gene–gene correlations, fold-change concordance, descriptive statistics).
- Applied the Kolmogorov–Smirnov test to compare gene-expression distribution shifts across datasets, identifying genes driving major between-species differences as interpretable targets for downstream analysis.
- Evaluated the feasibility of integrating DrugMatrix and Open TG-GATEs by organizing raw microarray data into analysis-ready structures and comparing cross-platform expression distributions after unified RMA preprocessing.

Summer Program for Undergraduate Research (SPUR), University of Utah May 2024 – Aug. 2024
Summer Undergraduate Researcher, advised by Prof. Andy Hong and Prof. Katherine E. Isaacs Salt Lake City, UT

- Examined how environmental factors relate to outdoor fall injuries among older adults by linking national EMS (NEMSIS) injury records with ZIP-code-level environmental data and fitting a zero-inflated negative binomial model.
- Mapped the geographic distribution of fall incidents and used R and Tableau to visualize temporal trends, spatial patterns, and demographic differences in fall risk; results were published in *RANGE: Undergraduate Research Journal*.

Publications & Presentations

Journal Article

[1] **Chen, Y.**, Hong, A., & Isaacs, K. (2024). Environmental determinants of outdoor fall injuries among older adults: A national study. *RANGE: Undergraduate Research Journal*. [Online article]
Also presented as a poster at the University of Utah SPUR Undergraduate Research Symposium, 2024.

Conference Papers & Posters

[2] **Chen, Y.**, Cong, G., Lim, S.-H., Chao, F., & Auerbach, S. S. (2026, March). *Statistical evaluation of DrugMatrix and Open TG-GATEs for construction of an integrated toxicogenomics resource* [Poster, Abstract No. 4385]. Society of Toxicology (SOT) Annual Meeting, San Diego, CA.

[3] Chen, S., **Chen, Y.**, Xu, T., Lee, C. T., & Sun, X. (2025, November). *From family context to achievement in diverse adolescents: Uncovering hidden family experience patterns with unsupervised machine learning* [Paper presentation]. National Council on Family Relations (NCFR) Annual Conference, Baltimore, MD.

[4] **Chen, Y.**, Cong, G., & Lim, S.-H. (2025, July). *Toxicogenomic variation in cross species and cross mode studies* [Poster]. Oak Ridge National Laboratory PCIP Poster Session, Oak Ridge, TN.

Selected Projects

Normalizing Flows for Gravitational-Wave Parameter Estimation Spring 2025
Collaborative course project, AST 8581: Big Data Analytics in Astrophysics, University of Minnesota [GitHub]

- Explored likelihood-free Bayesian parameter estimation for gravitational-wave signals: time-frequency representations of LIGO Hanford and Livingston data were encoded by a similarity embedding network (VICReg objective) and used as context for a masked autoregressive normalizing flow.
- Contributed flow data preparation, training and testing components, and diagnostic visualizations of learned representations and parameter estimates.

Cross-Database Integration of Rat Liver In Vivo Toxicogenomics Data Spring 2026
Statistics capstone, University of Minnesota

- Evaluated systematic batch differences between DrugMatrix and Open TG-GATEs rat liver gene-expression data, using PCA, distributional analysis, and a control-anchor correction strategy to make the data more comparable without erasing biologically meaningful signals.

Honors & Awards

- **Buehler Memorial Undergraduate Award** ($1,500), School of Statistics, University of Minnesota Sep. 2025
  Awarded annually to 3–5 Statistics undergraduates for excellence in statistical modeling and technical communication.
- **CLA Waller Scholarship** ($2,000), University of Minnesota May 2025
- **First Place, Undergraduate Category**, ASA Twin Cities Chapter Spring Poster Competition ($150) Apr. 2025
  For the poster "Environmental Determinants of Outdoor Fall Injuries among Older Adults: A National Study."
- **Undergraduate Research Opportunities Program (UROP) Award** ($900), University of Minnesota Nov. 2024
- **Dean's List**, University of Minnesota 2023 – 2025
- **Third Prize, Chongqing Region**, National College Student Statistical Modeling Competition, China May 2023
- **Merit Scholarship**, Southwest University of Political Science and Law Mar. 2022

Outreach & Activities

Volunteer Community Math and English Tutoring Oct. 2021 – Apr. 2022

- Provided individualized mathematics and English tutoring to children in rural communities through weekly online sessions, adapting explanations and activities to each student's needs.

Midwest Undergraduate Data Analytics Competition (MUDAC) Apr. 2024 & Apr. 2025

- 2025: Conducted text-based sentiment analysis of call-record notes from Big Brothers Big Sisters of Minnesota using Python and NLTK to identify themes and emotional signals.
- 2024: Investigated causes of missing agricultural data across Minnesota counties and built geographic and temporal visualizations of statewide agricultural indicators in Tableau.

Oak Ridge Computing Academy (ORCA): HPC System Administration Training, ORNL Jul. 2024

- Week-long hands-on training at the Oak Ridge Leadership Computing Facility on cluster setup, virtualized HPC nodes, and networking.

Skills

Programming & analysis: Python, R, Stata, SPSS · **Machine learning:** PyTorch, scikit-learn, pandas
Visualization: Tableau, ggplot2, matplotlib · **Tools:** Git, LaTeX, MongoDB, Linux